WORKED SOLUTION

# STEP II 1987, Question 13

Mechanics • Oblique Impacts with Walls

**STEP II 1987, Question 13**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q13

## The key idea

A smooth cushion can only change the velocity component perpendicular to it. That component is reversed and multiplied by $e$, and the component along the cushion is unchanged. Two adjacent cushions act on the two different components, so each gets reversed and scaled by $e$ once. For the closed paths, we follow the puck from cushion to cushion with coordinates and choose the one free quantity, the initial slope, so that the puck comes back to where it started.

## Solution

**Model.** The puck is a particle sliding on a smooth horizontal table with smooth cushions, so nothing acts on it between impacts and it moves in straight lines at constant velocity. At an impact the cushion's impulse is perpendicular to the cushion (it is smooth), so the parallel component of velocity is unchanged. By Newton's law of restitution the perpendicular component is reversed and multiplied by $e$. We take $0 < e \leq 1$.

Put axes along two sides of the table, so that it is $0 \leq x \leq L$, $0 \leq y \leq B$.

### Part (i): two adjacent cushions

Let the puck's velocity be $(u, v)$. Adjacent cushions are one cushion of the form $x = \text{const}$ and one of the form $y = \text{const}$, hit in either order. Hitting an $x$-cushion sends $(u, v) \mapsto (-eu, v)$, and hitting a $y$-cushion sends $(u, v) \mapsto (u, -ev)$. After one of each, in either order, the velocity is

$$(-eu,\ -ev) = -e\,(u, v).$$

This is a scalar multiple of the original velocity, so the final path is parallel to the original one, as required. The puck is now travelling in the opposite sense, with its speed reduced by a factor $e$.

**Part (ii): a closed path from a cushion**

Let the puck rest at $P = (a, 0)$ on the cushion $y = 0$, with $a : (L - a) = z : 1$, so that

$$a = \frac{zL}{1+z}, \qquad L - a = \frac{L}{1+z}, \qquad 0 < a < L.$$

Looking down on the table with $x$ to the right and $y$ upwards, going clockwise from the cushion $y = 0$ we meet $x = 0$, then $y = B$, then $x = L$. If the ratio is measured from the other end, replace $z$ by $1/z$; the argument is unchanged. Hit the puck with velocity $(-u, v)$, where $u, v > 0$, and write $k = v/u$ for the slope. The velocities on the four legs are

$$(-u, v) \xrightarrow{x=0} (eu, v) \xrightarrow{y=B} (eu, -ev) \xrightarrow{x=L} (-e^2u, -ev) \xrightarrow{y=0} \cdots$$

- Leg 1, from $(a, 0)$ to $x = 0$: it takes horizontal distance $a$ and rises $ak$, so it hits the left cushion at $W_1 = (0,\, ak)$.
- Leg 2, with slope $v/(eu) = k/e$: it rises $B - ak$ and so moves across $e(B - ak)/k$. It meets the top at $W_2 = \big(e(B/k - a),\, B\big)$.
- Leg 3, with slope $-k$: it moves across $L - x_{W_2}$ and falls $k(L - x_{W_2})$, so it meets the right cushion at height $y_{W_3} = B - kL + e(B - ak)$.
- Leg 4, with slope $k/e$: it falls $y_{W_3}$ and moves back $e\,y_{W_3}/k$, reaching $y = 0$ at

$$x_{\text{end}} = L - \frac{e\,y_{W_3}}{k} = (1+e)L + e^2a - \frac{e(1+e)B}{k}.$$

We want $x_{\text{end}} = a$. That gives $e(1+e)B/k = (1+e)L - (1-e^2)a = (1+e)\big(L - (1-e)a\big)$, so

$$k = \frac{eB}{L - (1-e)a} = \frac{B}{L} \cdot \frac{e(1+z)}{1+ez}.$$

This is a positive number for every $z > 0$, since $L - (1-e)a > L - a > 0$.

> **Ask yourself** Have we finished? We assumed the puck meets the cushions in the order left, top, right, bottom.

No. We still have to check that each impact point really lies on the side we assumed. Write $D = L - (1-e)a = L - a + ea$, so $k = eB/D$. Then

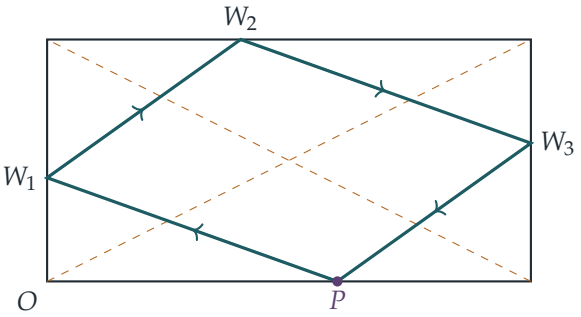
$$y_{W_1} = ak = \frac{eaB}{D},$$

$$x_{W_2} = e\Big(\frac{D}{e} - a\Big) = L - a,$$

$$y_{W_3} = B(1+e) - k(L + ea) = \frac{B\big[(1+e)(L - a + ea) - eL - e^2a\big]}{D} = \frac{B(L-a)}{D}.$$

Since $0 < a < L$ and $e > 0$ we have $0 < ea < D$ and $0 < L - a < D$. So $0 < y_{W_1} < B$, $0 < x_{W_2} < L$ and $0 < y_{W_3} < B$. Each impact is on the open side we assumed, and no corner is hit. So for every $z$, the shot with slope $k$ above goes left, top, right and comes back to $P$.

By part (i), legs 1 and 3 are parallel and so are legs 2 and 4, so the closed path is a parallelogram $PW_1W_2W_3$ inscribed in the table.




The dashed lines are the diagonals. The path drawn has $L = 2B$, $e = \frac{1}{2}$ and $z = 1.5$.

#### Part (iii): two shots from any interior point

*Every such shot is one of the part (ii) paths.* Suppose the puck starts at an interior point $Q$ with velocity $\mathbf{w}$, bounces at $V_1, V_2, V_3, V_4$ on the four cushions in clockwise order, and comes back to $Q$. Apply part (i) twice: after four bounces the velocity is $(-e)^2\mathbf{w} = e^2\mathbf{w}$, so it points the same way as the original. The last leg $V_4Q$ and the first leg $QV_1$ therefore lie on the same straight line, in the same direction. So $V_4 \to Q \to V_1$ is one straight leg, and the puck's path is a closed path that starts on a cushion at $V_4$, bounces off the other three clockwise and returns to $V_4$. The working in part (ii) shows that the slope of such a path is fixed by its starting point. Each such path has one vertex on each side, so all of them are the parallelograms $\Pi_a$ of part (ii), for $0 < a < L$, and $Q$ lies on a side of $\Pi_a$. Conversely, if $Q$ lies on a side of some $\Pi_a$, shooting from $Q$ along that side in the direction of travel works. So we must count the pairs (parallelogram, side) through $Q$.

*How the sides sweep the table as $z$ varies.* As $z$ runs from 0 to $\infty$, $a$ runs from 0 to $L$. From the formulae above, $y_{W_1}$ increases from 0 to $B$, $x_{W_2} = L - a$ decreases from $L$ to 0, and $y_{W_3} = B/\big(1 + ea/(L-a)\big)$ decreases from $B$ to 0. So $\Pi_a$ starts as the diagonal through $O$ (at $a \to 0$ the vertices crowd into the corners $O$ and $(L, B)$) and ends as the other diagonal. Now look at one side, say $PW_1$. It cuts off the triangle with legs $a$ along the bottom and $y_{W_1}$ up the left side at the corner $O$. Both legs increase strictly with $a$, so these triangles are strictly nested and their hypotenuses never cross. They grow continuously from nothing to the half of the table below the diagonal from $(L, 0)$ to $(0, B)$. Precisely, if $Q = (q_1, q_2)$ lies in that half, the function

$$f(a) = \frac{q_1}{a} + \frac{q_2}{y_{W_1}(a)}$$

is continuous and strictly decreasing. It tends to $\infty$ as $a \to 0$ and to $q_1/L + q_2/B < 1$ as $a \to L$, so $f(a) = 1$ (meaning $Q$ is on $PW_1$) for exactly one $a$. If $Q$ is not in that half, $f(a) > 1$ for every $a$ and $PW_1$ never passes through $Q$. The same argument applies to each side:

- $PW_1$ sweeps the half-table below the diagonal from $(L, 0)$ to $(0, B)$ once;
- $W_2W_3$ sweeps the half-table above that diagonal once;
- $W_1W_2$ sweeps the half-table above the diagonal from $O$ to $(L, B)$ once;
- $W_3P$ sweeps the half-table below that diagonal once.

A point $Q$ away from the cushions, and not on a diagonal, lies on exactly one side of each diagonal. So exactly two sides, belonging to two different parallelograms, pass through it. Those are the two shots: each one sends the puck along that side, round all four cushions clockwise and back to $Q$. (If $Q$ is on a diagonal, one of the two shots degenerates into a shot into a corner.)

## Checks

- With $e = 1$ the slope is $k = B/L$ for every $z$, so all the sides are parallel to the diagonals. That is what the usual mirror-image unfolding gives for perfectly elastic cushions.
- Take $L = 2$, $B = 1$, $e = \frac{1}{2}$, $z = 1$: then $a = 1$, $k = \frac{1}{3}$, and the impacts are at $(0, \frac{1}{3})$, $(1, 1)$ and $(2, \frac{2}{3})$. The last leg has $\mathrm{d}x/\mathrm{d}y = e/k = \frac{3}{2}$, so falling $\frac{2}{3}$ moves it back 1, to $(1, 0) = P$.
- In a numerical simulation that steps the puck from wall to wall, all three parts held: random tables, values of $e$ and values of $z$ gave the predicted impact points, and random interior points each lay on exactly two of the closed paths.

## Common slips

- Using the mirror-image (unfolding) trick: with $e < 1$ the angle of reflection is not equal to the angle of incidence, so the trick fails.
- Finding $k$ from $x_{\text{end}} = a$ and not checking that each impact lands on the intended side.
- In part (iii), assuming the shot from $Q$ must start along a particular side. The part (i) argument is what joins the first and last legs into one straight leg.

## Beyond STEP

Paths like these are *billiards*. With $e = 1$ they are a whole subject of their own, studied with reflections and unfoldings and leading into dynamical systems. Here the cushions lose energy, and the map "velocity before one round $\mapsto$ velocity after" is a pure scaling by $e^2$. That is why closed orbits exist for every $e$, even though the puck slows down on each lap. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers impulse and restitution carefully. Siklos, *Advanced Problems in Mathematics*, has more problems in the same spirit.

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